Candidate Number

Candidate Name____________________________

INTERNATIONAL ENGLISH LANGUAGE TESTING SYSTEM

Academic Reading

Additional materials:

Answer sheet for Listening and Reading
Time 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this question paper until you are told to do so.
Write your name and candidate number in the spaces at the top of this page. Read the instructions for each part of the paper carefully.
Answer all the questions.
Write your answers on the answer sheet. Use a pencil.
You must complete the answer sheet within the time limit.
At the end of the test, hand in both this question paper and your answer sheet.

INFORMATION FOR CANDIDATES

There are **40** questions on this question paper.
Each question carries one mark.

READING PASSAGE 1

*You should spend about 20 minutes on Questions **1–13**, which are based on Reading Passage 1 below.*

Effect and Cause

A chance finding by a Swiss research team explains an event that happened centuries ago around Lake Geneva

In the sixth century Gregory of Tours, a chronicler of the Germanic people known as the Rhone River spills into Lake Geneva. He wrote of a big rockfall in the year 563 AD in the vicinity of a place called Tauredunum. The debris plunged into the river, and a great mass of water overwhelmed with a sudden and violent flood all that was demolished, and several people inside the city walls of Geneva were killed. Historians and scientists have long believed that Gregory and another chronicler, Marius of Avenches, who told a similar tale, were describing a tsunami that raced across the lake. But there has not been any direct evidence of it until now.

Researchers at the University of Geneva now say they have found that evidence, in the form of a large deposit of sediment in the middle of the lake. In a study published in the journal Nature Geoscience, they also propose the sequence of events that caused the deadly surge. The researchers think that large boulder crashed down onto soft sediments which had accumulated at the river mouth because of the slowing of the river's flow when it enters the lake. These sediments formed an underwater delta that had several canyon-like channels. When the falling rocks hit the delta, they destabilized the sediments and caused the canyons to collapse. It was this collapse that created the tsunami. The sediments from this collapse would have been propelled towards the lake's centre.

Guy Simpson, a lecturer in the University of Geneva's Department of Geology and Paleontology, says the thick layer of sediment, which has the same curved shape as a lens, lies more than 305 metres down in the deepest part of the lake, and was found largely by chance. Katrina Kremer, a University of Geneva doctoral student and the study's lead author, had been conducting seismic soundings searching for thin sediment layers that might be evidence of major floods that had taken place in pre-historic times, long before the event described by Gregory of Tours. 'But we came across this enormous deposit,' Simpson says. 'We didn't know straight away that it was the deposit that caused the [sixth-century] tsunami. But it was a jumbled mess of sediment. It was quite obvious that it was deposited rapidly.'

The researchers then took samples of the sediments and used carbon-dating techniques on remains of leaves and other organic matter they found to determine when the deposit formed. This narrowed the range to a period between the late fourth century and the early seventh century. Other than the rockfall, there is no record of any special event during that period, Simpson says.

The researchers estimated that the deposit, which is at least 9.6 kilometres long by 4.8 kilometres wide, and averages about five metres thick, contains more than 248 million cubic metres of material. They ran multiple computer simulations showing that the collapse of that much sediment at the mouth of the Rhone would have caused a tsunami with an estimated height of 7.9 metres at Geneva -where it would have arrived in about 70 minutes. The rockfall itself may have been set off by a major earthquake, as some scientists have speculated.

Lake tsunamis, although unusual, are not unknown, says Richard Schweickert, an Emeritus Professor of Geology at the University of Nevada in Reno, in the United States. He cites evidence that the collapse of part of the shoreline of Lake Tahoe in northern California within the past 20,000 years caused a tsunami with wave heights of about 30 metres. There are two faults under the lake that could have caused an earthquake, he says and that the collapse of the Rhone delta sediments, as calculated by the Swiss researchers, would certainly be capable of moving a large amount of material into the lake. 'He suggests that the findings could be corroborated by careful mapping of the shoreline to look for unusual deposits or erosion left behind by the giant waves.

Simpson says the Rhone delta sediments might collapse again, perhaps from an earthquake or even their own weight. In the sixth century, Geneva was a small community, mostly behind walls on a hill, whereas today it is home to international organisations and about 200,000 people, many living in low-lying areas near the water, Testing the stability of nearby slopes, and creating more detailed models of how a tsunami could affect Geneva today, would provide a more accurate assessment of whether or not this is something the lakeside city should be concerned about.

Most tsunamis occur in oceans and are generated by earthquakes. However, the study is a reminder that even a landlocked nation like Switzerland is not immune to catastrophic waves.

Questions 1 - 7

Do the following statements agree with the information given in Reading Passage 1?
In boxes 1-7 on your answer sheet, write

TRUE *if the statement agrees with the information*
FALSE *if the statement contradicts with the information*
NOT GIVEN *if there is no information on this*

1. According to Gregory of Tours. The landslide which caused the flood happened near Tauredunum.

2. The city of Geneva was undamaged by the tsunami that Gregory described.

3. The work of Marius of Avenches supported the idea that there was a tsunami.

4. The rocks which fell into the delta were very hard and dense.

5. Richard Schweickert has published studies on lake tsunamis that have occurred in countries.

6. The shoreline of Lake Tahoe has remained unchanged for 20,000 years.

7. Parts of the population of Geneva now live closer to the lake than was the case in the sixth century.

Questions 8 - 13

Complete the flow-chart below.

Choose ***ONE WORD AND/OR A NUMBER*** *from the passage for each answer.*
Write your answers in boxes 8-13 on your answer sheet.

The evidence for a tsunami in Lake Geneva

Katrina Kremer was looking for indications of pre-historic **8** ……………….. that had occurred before the sixth-century event.
⬇
Instead her research team found a large lens-shaped deposit.
⬇
Researchers extracted samples from the deposit and dated organic material such as **9** ………………..
⬇
Carbon-dating confirmed that the deposit was present in the lake by the beginning of the **10** ……………….. Century
⬇
The volume of the deposit was then **11** ……………….. by the scientists
⬇
The researchers established the approximate **12** ……………….. and speed of the tsunami through extensive computer modelling
⬇
A likely cause of the rockfall was a significant **13** ……………….

READING PASSAGE 2

You should spend about 20 minutes on Questions 14–26, which are based on Reading Passage 2 below.

How do plants talk to each other?

Research shows that plants communicate and interact with each other in surprisingly subtle and sophisticated ways.

A In 1983, plant scientists Jack Schultz and Ian Baldwin reported that undamaged young maple trees increased their defense systems when exposed to maples that had been damaged by plant-eating (herbivorous) insects. The injured trees, they suggested, were alerting neighbors to the presence of a predator by releasing chemical signals into the air. But the plant research community did not accept this. The results were difficult to replicate, critics pointed out. Many also questioned how a trait could be evolutionarily stable if it benefited neighboring plants but not the plant releasing the signal. By the late 1980s, most ecologists felt that Schultz and Baldwin's ideas had been discredited.

B A decade later, however, a number of more carefully designed experiments began to yield convincing indications to the contrary. In 2000, evolutionary ecologist Richard Karban showed that wild tobacco plants became resistant to herbivores when grown in close proximity to sagebrush plants whose leaves had been damaged by cutting. This change appeared to be in response to chemicals – known as volatile organic compounds or VOCs – released by the sagebrush plants. Other researchers soon reported similar VOC-induced defense responses in several other plants, including lima bean, broad bean, barley, and corn. And in 2006, Karban showed that VOCs released by damaged sagebrush induce herbivore resistance in plants growing at distances of up to 60 cm, well within the range of sagebrush neighbors in nature.

C But the question still remains: 'Why should a plant waste valuable resources on a function which has no obvious advantage for it?' One hypothesis is that external communication channels are merely an extension of within-plant signaling. In sagebrush, lima bean, and poplar, VOCs released from damaged parts of a plant induce resistance in intact sections of the same plant, suggesting that each individual plant uses the signals to coordinate its own physiological responses to protect itself. Karban agrees, saying, 'The interplant signaling we see may be a result of plants co-opting that process.' Alternatively, VOC-based signaling between plants may have been favored because it enhances the 'extended fitness' of the sender by aiding related plants of the same species: a strategy known as kin selection.

D Over the past few years, a team led by Ariel Novoplansky of Ben-Gurion University of the Negev in Israel seems to have found proof that distress signals can be passed through the plants' roots. They planted garden pea plants in rows and subjected the first in each row to conditions similar to those experienced in a drought. They then evaluated the response by measuring the
microscopic holes on leaves, known as pores, which react when there is a shortage of water. After fifteen minutes, the stressed plant was seen to be closing its pores, followed by all of its neighbors, one by one. Importantly, in a control setup where root contact between neighboring plants was blocked, pores stayed open. Meanwhile, David Johnson's team at the University of Aberdeen in Scotland have been studying the labyrinths of hair-like fungi that occur around the roots of most plants. These fungi are involved in an important two-way relationship: in exchange for sugars, they provide plants with much-needed phosphorus and nitrogen. Research in which broad bean plants were infested with aphids – small herbivorous insects – revealed that these networks also served as a channel for warning neighboring plants of the infestation.

E Monica Gagliano of the University of Western Australia believes that plants may even use sounds to alert their neighbors. In one study, she demonstrated that chili plants grown next to fennel plants developed more quickly than seedlings grown with other chili plants. Gagliano and her colleagues suspect the chili plants were compensating for the presence of the fennel, which is known to release chemicals that inhibit the growth of other plants. Remarkably, however, known plant communication pathways – airborne volatiles, root contact, and common fungal networks – were blocked. The results begged for an alternative explanation. 'We think this other channel of communication might be acoustic,' said Gagliano. But behavioral ecologist Carel ten Cate of the University of Leiden in the Netherlands points out that taking advantage of such benefits would require sensory mechanisms yet to be described in plants. Despite widespread skepticism, Gagliano has received some encouragement. Richard Karban, for example, is cautiously enthusiastic. 'Whether or not the explanation she favors is the right one,' he says, 'I think that she's gotten results that, as a field, we need to come to grips with.'

F Researchers believe that knowledge of this phenomenon could eventually be applied to agriculture, and lead to the cultivation of hardier crops. But all agree that there is still much work to be done. 'Applying our limited knowledge [of plant-communication mechanisms] to agriculture is a big jump,' says Johnson, 'but it is definitely on the horizon.' Nevertheless, one message has emerged loud and clear from those studying this new realm of botanical interaction: despite not possessing eyes, ears, or a nervous system, plants are anything but uncommunicative. ‘During my PhD [in the late 1980s], all this stuff was considered very weird,’ recalls Ariel Novoplansky. ‘Today, there is no doubt we now recognise that plants are capable of some very sophisticated exchanges of information with other plants. This idea is not strange anymore.’

Questions 14–19

*Reading Passage 2 has six paragraphs, **A-F.***
Choose the correct heading for each paragraph from the list of headings below.
*Write the correct number, i-viii, in boxes **14-20** on your answer sheet.*

List of Headings

i. Sending messages underground

ii. Potential advantages for farmers

iii. The widespread rejection of an idea

iv. One species accelerating growth in another

v. Some obstacles to plant communication

vi. Evidence leading to renewed belief in a theory

vii. New technologies which helped researchers

viii. A suggested benefit to the sender of plant communication

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14. Paragraph **A**

15. Paragraph **B**

16. Paragraph **C**

17. Paragraph **D**

18. Paragraph **E**

19. Paragraph **F**

Questions 20–23

Complete the summary below.
*Choose **ONE WORD ONLY** from the passage for each answer.*
*Write your answers in boxes **20 - 23** on your answer sheet.*

Plant communication research in Israel and Scotland

Plants may send messages via their **20 ______** according to Ariel Novoplansky of Ben-Gurion University in Israel. In one study, Novoplansky's team planted rows of garden peas, and created conditions like a **21 ______** around the first plant in each row. The plants responded by closing the tiny holes on their leaf surfaces, known as pores. They then appeared to signal to plants nearby, causing them to respond in a similar way.

Research at the University of Aberdeen in Scotland, meanwhile, has focused on the networks of **22 ______** which connect the plants beneath the soil. These structures, which extract sugars from the plants and supply them with phosphorus and nitrogen, also seem to play a key role in plant communication. Experiments involving broad bean plants and plant-eating insects called **23 ______** have also produced some interesting results.

Questions 24–26

Look at the following researchers (Questions 24 - 26) and the list of statements below. Match each researcher with the correct statement, A–E.
*Write the correct letter, A–E, in boxes **24 - 26** on your answer sheet.*

24. It is likely that research into plant communication will one day help to improve our food supply.
25. Plant communication could occur as a consequence of internal mechanisms intended to help a single plant.
26. Attitudes towards plant communication have changed greatly in recent decades.

List of Researchers

A. Richard Karban
B. Ariel Novoplansky
C. David Johnson
D. Monica Gagliano
E. Carel ten Cate

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READING PASSAGE 3

You should spend about 20 minutes on Questions 27–40, which are based on Reading Passage 3 on pages 10 and 11.

The future of the world's languages

Paul Bignell considers why half of the world's languages are in serious danger of dying out

Of the world's 6,500 living languages, around half are expected to die out by the end of the century. Just 11 are spoken by more than half the Earth's population, so it is little wonder that those used by only a few are being left behind as the world becomes a more homogenous place. In short, 95 per cent of the world's languages are spoken by only five per cent of its population – a remarkable level of linguistic diversity stored in tiny pockets of speakers around the world. However, Dr Mark Turin, a Cambridge University academic, has launched the World Oral Literature Project (WOLP), which aims to pull thousands of languages back from the brink of extinction.

Turin hopes to encourage small communities to collaborate with anthropologists around the world to record what he calls 'oral literature' through video cameras and voice recorders, by awarding grants from a £30,000 pot that the project has secured this year. For many small communities, the oral tradition is at the heart of their culture. The stories they tell are creative works as well as communicative. The idea is to collate this literature in a digital archive that can be accessed on demand and make the building blocks of lost cultures readily available. Unlike languages with celebrated written traditions, such as Ancient Greek, few small communities have made recordings of their own languages. Until now.

The project suggested itself when Turin was teaching in Nepal. He recalls that he had wanted to study for a PhD in endangered languages and, while discussing it with his professor at Cambridge University, was drawn to a map on his tutor's wall. The map was full of coloured pins which represented the world's languages that were completely undocumented. Turin picked out a 'pin'. It happened to belong to the Thangmi tribe, a community living near Kathmandu, the capital of Nepal. And so that is where Turin went. 'Many of the choices that anthropologists and linguists who work on these traditional field-work projects take are quite random,' he admits.

As part of his work with the Thangmi community, Turin began to record the language he was hearing. He realised that not only was this language and its culture entirely undocumented, it was known to few outside the tiny community. 'I started writing 1,000 pages of grammar in English, but I realised it wasn't enough. Unfortunately, it simply wasn't going to work as something for the community. So then I produced this trilingual word list in Thangmi, Nepali and English.'

In short, it was the first ever publication of the Thangmi language. That small dictionary is still sold in local schools, and used as part of a wider cultural regeneration process to educate children about their heritage and language. The task is no small undertaking: Nepal is a country of massive ethnic and linguistic diversity, home to 100 languages from four different language families. What's more, ever fewer ethnic Thangmi speak the Thangmi language. Many of the community members now speak Nepali, the national language taught in schools and spread by the media.

Despite Turin's enthusiasm for his subject, he is baffled by many of his fellow linguists' refusal to engage with the issue he is working on. 'There are more linguists in universities around the world than there are spoken languages – but most of them aren't working on the issue of their disappearance. To me it's amazing that in this day and age, we still have an entirely incomplete image of the world's linguistic diversity. People do PhDs on the apostrophe in French, yet we still don't know how many languages are spoken. When a language becomes endangered, so too does a cultural world view. We want to engage with indigenous people to document their myths and folklore as well as their language.'

Yet, despite the struggles facing initiatives such as WOLP, there are historical examples that point to the possibility that language restoration is no mere academic pipe dream. The revival of a form of Hebrew in the 19th century is often cited as one of the best proofs that languages long dead, belonging to small communities, can be resurrected and embraced by a large number of people. Today, it is spoken by more than seven million people.

Dr Turin believes that the fate of the world's endangered languages is not sealed, and globalisation is not necessarily the perpetrator of evil it is often presented to be. 'It's a paradox really: on the one hand globalisation and rapid socio-economic change are the things that are eroding and challenging diversity. But on the other, globalisation is providing us with new and very exciting tools and facilities to get to places, to document those things that globalisation is eroding. Also, the communities that are affected by change are excited by what globalisation has to offer.'

In the meantime, the race is on to collect and protect as many of the languages as possible, so that future generations can continue their traditions and have a sense of identity. And it certainly is a race: Turin knows his project's limits and believes it inevitable that a large number of those languages will die out. 'We have to be realistic. A project like ours is in no position to, and was not designed to keep languages alive. The only people who can help languages survive are the people in those small communities themselves. They need to be reminded that it's good to speak their own language. Becoming modern doesn't mean you have to stop using your language. I think it's important they understand that.'

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Questions 27–31

*Complete the summary using the list of words, **A–J**, below.*
*Write the correct letter, **A–J**, in boxes 27–31 on your answer sheet.*

The world's languages

Approximately half of the world's languages are expected to die out by the end of this century. As a result of globalisation, many of the languages that are spoken by a **27** ______ of people will become extinct. However, Dr Mark Turin has started a project to prevent the **28** ______ of these languages. The project provides communities around the world with **29** ______ to enable them to record their oral traditions. Oral traditions have great cultural **30** ______ in these communities. However, an important **31** ______ between the languages of these small communities, and those of cultures with celebrated written traditions, is that most of the former have not been recorded before now.

A. adaptation	**B** significance	**C** disappearance
D similarity	**E** globalization	**F** difference
G minority	**H** education	**I** variety
J funding		

Questions 32–35

Do the following statements agree with the claims of the writer in Reading Passage 3? In boxes 32–35 on your answer sheet, write:

***YES** if the statement agrees with the claims of the writer*
***NO** if the statement contradicts the claims of the writer*
***NOT GIVEN** if it is impossible to say what the writer thinks about this*

32. Turin argues that anthropologists and linguists usually think carefully before selecting an area to research.

33. Turin concluded that the language spoken by the Thangmi community shared few similarities with other languages.

34. Turin realised that the 1000-page document he wrote was inappropriate for the Thangmi community.

35. Some Nepalese schools lack the resources to devote to language teaching.

Questions 36–40

*Choose the correct letter, **A, B, C or D.***
*Write the correct letter in boxes **36–40** on your answer sheet.*

36. Why does Turin refer to PhDs on the apostrophe in French?

A. to illustrate the limited scope of much linguistic research
B. to acknowledge how linguistic research has improved
C. to compare different approaches to the role of punctuation
D. to suggest how research contributes to a diverse cultural outlook

37. The main point Turin is making in the seventh paragraph is that

A. WOLP has a chance of achieving some success.
B. the difficulties facing WOLP have been underestimated.
C. academics are interested in certain types of endangered language.
D. the threat to endangered languages is greater now than in the 19th century.

38. What does Turin say about globalisation?

A. The forces driving globalisation are unclear.
B. Small communities tend to distrust globalisation.
C. Globalisation has both advantages and disadvantages.
D. People need to have a more critical view of globalisation.

39. How does Turin feel about WOLP's prospects?

A. He accepts that the project will be viewed differently by future generations.
B. He believes the project will not prevent some languages from disappearing.
C. He agrees that researchers involved in the project will need to work faster.
D. He hopes the project will enable communities to develop a new identity.

40. Turin would like to tell people in small communities that they

A. are entitled to expect more benefits from the WOLP project.
B. need to accept their language has a limited role in modern society.
C. should consider documenting aspects of their language themselves.
D. can keep up with developments in the world without losing their language.

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